Literature Review

Ze Yu Zhong

We begin by providing a broad overview of the literature thus far.

Empirical Motivation

A factor model for a $T \times N$ panel of time series X assumes that X is generated from lower dimensional "factor structure":

$$\begin{aligned} X_{it} &= \lambda_{it}' F_t + e_{it} \\ X_t &= \Lambda_t F_t + e_t \\ X &= F\Lambda' + \mathbf{e}. \end{aligned}$$

which correspond to the individual, time based, and whole panel representations respectively, and are equivalent.

There are broadly two motivations for the use of factor models in environments with both a large number of cross sectional units and large number of observations. The first is that factor models correspond to the "factor structure" model assumption, where it is implicitly assumed that there exists a small number of true latent factors which propagate throughout the data. Such a motivation is popular in economics for interpretative purposes, as these true latent factorsc an be interpreted as

Exact Factor Models

A factor model for a $T \times N$ panel of time series X assumes that X is generated from lower dimensional "factor structure":

$$\begin{aligned} X_{it} &= \lambda_{it}' F_t + e_{it} \\ X_t &= \Lambda_t F_t + e_t \\ X &= F\Lambda' + \mathbf{e}. \end{aligned}$$

which correspond to the individual, time based, and whole panel representations respectively, and are equivalent.

The first representation is useful for analysing idiosyncratic disturbances, the second is useful for analysing changes that are common across the entire cross section, and the third is useful for splitting the entire panel into different time based subsamples.

An exact factor model assumes that the idiosyncratic shocks e_{it} do not have any serial correlation or cross sectional correlation, a strict assumption that is unrealistic for empirical applications.

Approximate Factor Models

The *exact* factor model can be subsequently loosened to an *approximate* factor model, in the sense that the idiosyncratic shocks are now allowed to have a limited degree of serial and cross sectional correlation. A precise definition of this degree is provided by Bai and Ng ((2002)) and Bai ((2003)).

An approximate factor model has the *static* representation if the factors F_t can only affect X_t directly, and not through its lags:

$$X_t = \Lambda F_t + e_t$$

The above formulation is to be contrasted with the *dynamic* form of the dynamic factor model, where the fact that the factors themselves are allowed to evolve dynamically is made more explicit:

$$X_{it} = \lambda_i'(L) f_t + e_{it}$$

where $\lambda_i'(L) = (1 - \lambda_{1i}L - \cdots - \lambda_{is}L^s)$ is a vector of dynamic factor loadings or order s. Thus, the static factors F_t refers to space spanned by the leads and lags of the dynamic factors f_t. We focus on a finite s, in contrast to the separate strand of literature focused on *generalized* dynamic factor models which allows s to be infinite (see Forni et al. ((2000)) and later section for a short review).

In either the static or dynamic formulations, the factors themselves are allowed to evolve according to:

$$f_t = C(L)\epsilon_t$$

where ϵ_t are $q \times 1$ i.i.d. errors, which can be interpreted as the *primitive shocks* in an economic model. The literature which focuses on these primitive shocks refers to q as the number of *dynamic* factors.

The dynamic factor model with q *dynamic* factors can always be re-written as a static factor model with r *static* factors, where r is finite. Because F_t will contain the leads and lags of f_t, the general relationship $q \leq q(s+1) = r$ always holds (Bai and Ng ((2007))).

It is because of this relationship that the literature usually does not explicitly differentiate between the two, referring (somewhat confusingly) to both the Static Factor Model and Dynamic Factor Model as simply Dynamic Factor Models, with the former referred to as the *static representation* of the Dynamic Factor Model when distinction is necessary.

Empirically, it has been noted that forecasting approaches using either formulation perform rather similarly. However, for the purposes of interpretation such as the identification of primitive shocks, distinction is important.

Generalized Dynamic Factor Models

In a separate, but related literature, Forni et al. ((2000)) develop so the called *Generalized* Dynamic Factor Model (GDFM) in which s is explicitly allowed to $\to \infty$. Estimation and analysis of this model instead focuses on the spectral density of X.

However, this approach is noticeably more complicated, and is still subject to its own issues, such as increased computational intensity, and the necessity of formulating a "one-sided" filter in order to reliably estimate the factors towards the end of the sample for forecasting purposes.

It is perhaps due to these difficulties that in spite of the dynamic factors as estimated via spectral analysis have been shown to have good performance by its authors, that the applied forecasting literature has, at large, stuck to using traditional static factor models, usually as estimated via the principal components estimator.

Latent Factor Models

More recently, Lam, Yao, and Bathia ((2011)) develop and formulate the *Latent* Factor Model (LFM).

The key difference in this formulation is that serial correlation in the error term is explicitly allowed for, and in fact, *required* to achieve identification. Estimation of the factors and loadings is done via applying an eigendecomposition of the empirical autocovariance matrices of the data.

Lam and Yao ((2012)) subsequently develop the inferential theory for the number of *latent* factors.

Although there has been some work done with this model as a basis, (see Liu and Chen ((2017)), Liu and Chen ((2020)), Liu and Zhang ((2022))), in general, this has not proven to be a popular way to estimate factor models.

The precise relationship between this latent factor model with the existing literature is an unanswered question.

Structural Breaks in DFMs

The topic of structural breaks in the context of Dynamic Factor Models has received much attention recently.

Due to simplicity, almost all of the literature thus far has focused on the the Dynamic Factor Model as estimated (statically) via the PC estimator.

Small Changes in Factor Loadings

We first make a preliminary and necessary distinction between so called "small" and "large" breaks in dynamic factor models. Heuristically, the estimation of F_t via the principal components estimator is not affected asymptotically if the breaks in the factor loadings is "small". Bates et al. ((2013)) establish the conditions under which the changes in Λ can be ignored for the estimation of F_t. They consider the following formulation:

$$X_{it} = \lambda_{it}' F_t + e_{it},\ i = 1, \ldots, N, t = 1, \ldots, T.$$

where λ_{it} is now allowed to be time-varying. Define $\Lambda_t = (\lambda_{1t}, \ldots, \lambda_{Nt})'$ and let the initial factor loading Λ_0 be fixed. The factor loading at time t is then allowed to evolve over time according to:

$$\Lambda_t = \Lambda_0 + h_{NT}\xi_t$$

where h_{NT} is a deterministic scalar allowed to depend on N, T, $\xi_t = (\xi_{1t}, \ldots, \xi_{Nt})'$ is an $N \times r$ stochastic process which governs the dynamics of Λ_t.

Combining the above, we have:

$$X_t = \Lambda_0 F_t + h_{NT}\xi_t F_t + e_t.$$

Bates et al. ((2013)) treat $h_{NT}\xi_t F_t + e_t$ as the combined idiosyncratic error for the factor model, and develop conditions under which the combined idiosyncratic error does not affect the consistency of the PCA estimator $\widehat{F}$ in Bai and Ng ((2002)). Although a rigorous set of conditions on h_{NT} is provided, for brevity, we provide the three most empirically relevant cases as derived by Bates et al. ((2013))

Example 1 White Noise: h_{NT} are i.i.d. across all i, t and $E||\xi_{it}||^4 < \infty$. If $h_{NT} = O(1)$, then consistency holds.

Example 2 Random Walk: h_{NT} are random walk processes, i.e. $\xi_{it} = \sum_{s=1}^{t} \zeta_{is}$, where ζ_{is} are i.i.d. across all i, t and $E||\xi_{it}||^4 < \infty$. A sufficient condition for consistency is $h_{NT} = o(T^{-1/2})$. Heuristically, this means that as long as the factor loading innovations shrink sufficiently fast enough, then PCA will not be affected.

Example 3 Single Large Break: let $\delta_i \in \mathbb{R}^r$ be a shift parameter. Then assume that the factor loadings are subject to a one time shift:

$$\xi_{it} = \begin{cases} 0_{r\times 1} & t = 1, \ldots, \lfloor \pi_0 T \rfloor \\ \delta_i & t = \lfloor \pi_0 T \rfloor + 1, \ldots, T. \end{cases}$$

If $h_{NT} = O(1)$ *and* at most $O(N^{1/2})$ series undergo a break, then PCA is still consistent. Example 3 is perhaps the most directly relevant case in the analysis which follows. Heuristically, PCA is still consistent as long as the break affects a shrinking fraction of all series asymptotically, and the their sizes are fixed.

It is important to note that the above conditions are stated only for the consistency of F, and *not* for the number of factors, which in general *is* affected even by the presence of small breaks. It is for this reason that a more recent strand of literature has emerged, focusing on the detection of small breaks to facilitate accurate estimation of the number of factors can be accurately determined before and after possible *small* breaks (see Bai, Han, and Shi ((2020)) and Duan, Bai, and Han ((2021)) forthcoming).

In what follows, we focus on *large*, one time breaks (Example 3) in dynamic factor models.

Model Setup and Notation

We focus on the case of *large*, one time breaks in dynamic factor models. To begin, we consider the singular break case in the factor loadings within the static model notation as proposed by Han and Inoue ((2015)):

$$X_{it} = \begin{cases} F_{0,t}'\lambda_{0,i} + F_{1,t}'\lambda_{1,i} + e_{it} & \text{for} \quad t = 1, \ldots, \lfloor \tau T \rfloor \\ F_{0,t}'\lambda_{0,i} + F_{1,t}'\lambda_{2,i} + e_{it} & \text{for} \quad t = \lfloor \tau T \rfloor + 1, \ldots, T, \end{cases}$$

where $F_{0,t}$ is a $(r-q) \times 1$ vector of time invariant factors, $F_{1,t}$ is a $q \times 1$ dimensional factor whose loadings undergo structural change.

The matrix representation of the the previous setup is:

$$\begin{aligned} X &= \begin{bmatrix} F^{0,1}\Lambda_0' + F^{1,1}\Lambda_1' \\ F^{0,2}\Lambda_0' + F^{1,2}\Lambda_1' \end{bmatrix} + e \\ &= \begin{bmatrix} F^{0,1} & F^{1,1} & 0 \\ F^{0,2} & 0 & F^{1,2} \end{bmatrix} \begin{bmatrix} \Lambda_0 & \Lambda_1 & \Lambda_2 \end{bmatrix}' + e \\ &= \mathcal{F}\Theta + e. \end{aligned}$$

Note that the number of factors above is possible larger than r, and depends on the rank of $\lim_{N\to\infty} \Theta'\Theta/N$ or more succintly (and equivalently) the row rank of Θ. In general, $r \leq \operatorname{rank} \Theta'\Theta/N \leq r = q$ and $\operatorname{rank} \Theta$. Han and Inoue ((2015)) further classify breaks into 3 different types:

1. $rank(\Theta) = r + q_1$, Type 1 Break

The changes in $\lambda_{1,i} - \lambda_{2,i}$ need to be so idiosyncratic across i such that Λ_1 and Λ_2 are linearly independent. The type of breaks in loadings considered by Breitung and Eickmeier ((2011)) corresponds to a Type 1 Break.

2. $rank(\Theta) = r$, Type 2 Break

The column rank of $[\Lambda_1 : \Lambda_2]$ is q_1, so that there exists a $q \times q_1$ matrix Z_0 such that $\Lambda_2 = \Lambda_1 Z_0$.

A type 2 break implies that the "break" is simply a rotation or scaling of the existing loadings. The practical interpretation of a type 2 break is that *all* loadings change in a homogeneous way. It is remarkable that type 2 changes can be interpreted as either a change in the common dynamic factors, or a homogeneous change in all factor loadings - it is not possible to separately identify the two cases. From a practical perspective however, a change in the underlying factor dynamics is more plausible than all factor loadings changing in a homogeneous way, and as such, it is arguably more reasonable to interpret type 2 breaks as breaks in factor dynamics, rather than breaks in factor loadings.

Although Han and Inoue ((2015)) allow Z_0 is allowed to be singular, in which case there will be emerging (disappearing) factors after the break, we further distinguish the singular case, as it turns out that different tests can treat this in a distinctly different way.

Type 2a: Z_0 is singular

The singular transformation case is the most topical for practical applications, as people tend to be interested in emerging or disappearing factors. Note that this type of break is explicitly ruled out by assumption by Chen, Dolado, and Gonzalo ((2014)), though the authors remark that their test appears to have power against this type of break. Intuitively, from the perspective of changing factor loadings, a type 2a break where one dynamic factor disappears can be equivalently viewed as a break where *all* of its associated factor loadings change to 0.

Type 2b: Z_0 is non-singular

The non-singular case corresponds to the more general case where factors have their loadings changed in a homogeneous way across *all* series. Although ruled out by Han and Inoue ((2015))'s regularity assumption, this in fact more generally corresponds to the case where there is a break in the dynamic factors, such as a change in the dynamic factor loadings, or a change in factor variance.

3. $r < rank(\Theta) < r + q_1$, Type 3 Break

A type 3 break is simply any combination of both a type 1 and type 2 break, which arises out of technical necessity.

Chen, Dolado, and Gonzalo ((2014)), Han and Inoue ((2015)) and Baltagi, Kao, and Wang ((2017)) propose tests for structural breaks in the factor loadings. However, because breaks in the factor loadings and factor dynamics are in general not separately identifiable, they all assume, as an identification condition that the factors are are stationary before and after the break (have the same variance). Han and Inoue ((2015)) and Baltagi, Kao, and Wang ((2017)) both formulate Equivalent Representation Theorems, which state that PCA will estimate an equivalent factor model with time invariant loadings, but with all breaks in the loadings transmitted to the estimated pseudo factor space. Based on this, they formulate a est statistic based on the second moments process of the pseudo factors.

However, procedures based on the pseudo factors cannot be accurately stated to be a test against breaks in factor loadings, because breaks in the pseudo factors may be in fact due to any combination of the following:

1. breaks in the loadings and/or
2. breaks in the factor dynamics and/or
3. breaks in the factor innovations.

The identification assumption of stationary factors explicitly rules out points 2 and 3, even though the distinction and resulting interpretation in the differences between these cases are different. J. H. Stock and Watson ((2016)) in their review chapter on Dynamic Factor Models remarks that many of these tests are new, and may have power against breaks in the factor dynamics as well.

In our following simulation study, we investigate the performance and properties of these tests in various cases.

Ma and Su ((2018)) provides a methodology to identify breaks in the loadings without assuming stationary factors. However, their method does not allow for a differing number of factors across regimes. More importantly, their method, being motivated via the LASSO literature, requires the specification of hyperparameters, and is very sensitive to this arbitrary choice. Practically, and it is unclear how to choose these for outside of synthetic data, as results are very non-robust.

Disentangling Breaks in Factor Loadings and Breaks in Factor Dynamics

The below table summarises available tests.

Test	Type 1	Type 2a	Type 2b	Type 3	Notes
Breitung and Eickmeier ((2011))	Yes*	n/a	n/a	n/a	Only for single series i
Yamamoto and Tanaka ((2015))	Yes*	n/a	n/a	n/a	Only for single series i
Chen, Dolado, and Gonzalo ((2014))	Yes	Yes*	No	Yes*	Test statistic has power against Type 2a, but no theoretical result
Han and Inoue ((2015))	Yes	Yes	Yes	Yes	Cannot distinguish between different types
Baltagi, Kao, and Wang ((2017))	Yes	Yes	Yes	Yes	Cannot distinguish between different types
Baltagi, Kao, and Wang ((2021))	Yes	Yes	Yes	Yes	Cannot distinguish between different types

Test	Type 1	Type 2a	Type 2b	Type 3	Notes
Bai, Han, and Shi ((2020))	Yes	No	Yes	Yes	Cannot distinguish between different types
Duan, Bai, and Han ((2021))	Yes	No	Yes	Yes	Cannot distinguish between different types, extremely large trimming

Thus, ignoring the separability issue, the problem of disentangling the breaks in factor loadings and those in factor dynamics can be be reduced down to the same problem as distinguishing between the different *types* of breaks.

Our goal is to propose a methodology that is able to detect all types of breaks, as well as classify *which* type of break they are, in order to pin down the precise source of the break (at least up to the separability constraint).

Of particular note is the test proposed by Breitung and Eickmeier ((2011)) and by extension, Yamamoto and Tanaka ((2015)), as this does appear to legitimately correctly test for breaks in loadings. However, one crucial assumption for this test is that the true factors are correctly estimated (at least up to a rotation) in a first step procedure, though this is somewhat alleviated by Yamamoto and Tanaka ((2015)). Additionally, because Breitung and Eickmeier ((2011))'s test is formulated under the null hypothesis than *all* factor loadings are constant over time, it has the tendency to over reject the null hypothesis for a single series, if a fraction of N other series have breaks in their loadings. However, more importantly, Bai and Han ((2016)) note that this sort of test has power against small breaks, and as such their results may be misleading if one is interested in estimating large common breaks. Additionally, Breitung and Eickmeier ((2011))'s test is only formulated for a single series, and cannot be generalized to the entire panel without restrictive assumptions on the error terms.

Chen, Dolado, and Gonzalo ((2014)) provides the closest methodology is achieving our goal of disentangling the source of breaks - their method has power against type 1 and type 2a breaks and no power against type 2b breaks - however, the asymptotics of their statistic has only been formally established for type 1 breaks.

It is also remarkable that another possible way to disentangle the breaks follows the remark by Chen, Dolado, and Gonzalo ((2014)) - type 2b breaks do never augment the estimated factor space with more factors, and a method based on the estimated number of factors before and after the break compared to the full sample may be possible.

We do not pursue this method for two reasons. First, it has been noted that estimators of the number of factors are sensitive to even small breaks, and empirically, are often in disagreement with one another. Second, recall that both type 1 and type 2a breaks can affect the no. of factors, meaning that this strategy still cannot disentangle between breaks in factor loadings and factor dynamics. That is, all breaks which do not change the no. of pseudo factors are type 2 breaks (corresponding to breaks in factor dynamics), but not all type 2 breaks change the no. of pseudo factors. This limitation makes it ultimately unsuitable even when used in conjunction with any pre-existing method.

Proposed Procedure

We propose a two step procedure in order to identify which kind of break, and therefore disentangle the source of the break.

The intuition behind our procedure is motivated by the fact that a type 2 break affects *all* series in a homogeneous way. Noting that this sort of break can be detected via existing methods, in the absence of other sorts of breaks, it stands to reason then that a simple estimate of long run variance for each series, averaged over the entire cross section will yield an estimator for the factor variance.

As a first step procedure, we propose to "standardize" the data by the inverse of this estimate of the factor variance. The resulting data should therefore not exhibit any "breaks" induced by changes in factor dynamics.

This adjusted data then satisfies the piecewise stationary factors identification condition (which was unreasonable) that existing methods assume, and thus these can then used as the second step in a 2 stage procedure to disentangle

the breaks.

The asymptotic inferential validity of the proposed procedure are yet to be established.

Differentiating between Type 1 and Type 2 breaks

Our proposed procedure is in essence a way to "undo" the common rotational type break implied by type 2 breaks as a first stage procedure. However, the treatment between the type 2a break and type 2b break are notably quite different.

Type 2a breaks imply that the rotational change is singular, and therefore there are disappearing/emerging factors. This break is useful, as it is the only mechanism in which certain types of factor dynamic changes can be incorporated, namely appearance/disappearance of lags, or an introduction of a new factor entirely.

However, this type of break also implies that the rotation brings about completely new information, and as such, "undoing" this sort of change is very difficult.

In contrast, the type 2b break is much easier, and seems to be working for now.

Simulation Setup

We consider 9 different simulation specifications to explore how different types of breaks are handled by different tests.

1. Null Case - no breaks

The first experiment contains no breaks and thus acts as a control.

$$X_{it} = F_t'\lambda_i + e_{it}$$

where F_t is a (3×1) vector of static factors, and each $\lambda_i \sim i.i.d.N(0_3, I_3)$.

2. Idiosyncratic Break in Factor Loadings (Type 1)

$$X_{it} = \begin{cases} F_t'\lambda_{1i} + e_{it} & \text{for} \quad t = 1, \ldots, \lfloor \tau T \rfloor \\ F_t'\lambda_{2i} + e_{it} & \text{for} \quad t = \lfloor \tau T \rfloor + 1, \ldots, T, \end{cases}$$

where $\lambda_{1i} \sim i.i.d.N(0, I_{N\times r})$, which changes to $\lambda_{2i} \sim i.i.d.N(0, I_{N\times r})$.

3. Break in Factor Dynamics (Type 2b)

$$\begin{aligned} X_{it} &= f_t'\lambda_i + e_{it}, \\ f_t &= \begin{cases} \rho_1 + f_{t-1} + \epsilon_t & \text{for} \quad t = 1, \ldots, \lfloor \tau T \rfloor \\ \rho_1 + f_{t-1} + 2\epsilon_t & \text{for} \quad t = \lfloor \tau T \rfloor + 1, \ldots, T, \end{cases} \end{aligned}$$

where the factor variance (via the innovations) undergoes a break. This corresponds to hetereoskedasticity in the underlying factors.

4. Break in Factor Loadings and Factor Dynamics (Type 3)

$$X_{it} = \begin{cases} F_t'\lambda_{1i} + e_{it} & \text{for} \quad t = 1, \ldots, \lfloor \tau T \rfloor \\ F_t'\lambda_{1i} + e_{it} & \text{for} \quad t = \lfloor \tau T \rfloor + 1, \ldots, T, \end{cases}$$

5. Common Break in Factor Loadings (Type 2b)

All factor loadings are assumed to undergo the *same* type of break. This is mathematically identical to a break in factor dynamics.

$$X_{it} = \begin{cases} F_t'\lambda_{1i} + e_{it} & \text{for} \quad t = 1, \ldots, \lfloor \tau T \rfloor \\ F_t'(2\lambda_{1i}) + e_{it} & \text{for} \quad t = \lfloor \tau T \rfloor + 1, \ldots, T, \end{cases}$$

in which the factor loadings are all multiplied by the same constant after the break.

6. Common Break in Idiosyncratic Error

$$X_{it} = \begin{cases} F_t'\lambda_i + e_{it} & \text{for} \quad t = 1, \ldots, \lfloor \tau T \rfloor \\ F_t'\lambda_i + 2e_{it} & \text{for} \quad t = \lfloor \tau T \rfloor + 1, \ldots, T, \end{cases}$$

where the "break" is now in the idiosyncratic error term.

7. Break in Factor Autocorrelation (Type 2b)

$$\begin{aligned} X_{it} &= f_t'\lambda_i + e_{it}, \\ f_t &= \begin{cases} \rho_1 + f_{t-1} + \epsilon_t & \text{for} \quad t = 1, \ldots, \lfloor \tau T \rfloor \\ \rho_2 + f_{t-1} + 2\epsilon_t & \text{for} \quad t = \lfloor \tau T \rfloor + 1, \ldots, T, \end{cases} \end{aligned}$$

where $\rho_1 = 0.7$, $\rho_2 = 0.3$. Clearly, a change in the autocorrelation coefficient of the factor process will also change the factor variance.

8. Appearance of Extra Factor Lag (Type 2a)

$$X_{it} = \begin{cases} F_t'\lambda_{1i} + e_{it} & \text{for} \quad t = 1, \ldots, \lfloor \tau T \rfloor \\ F_t'\lambda_{2i} + e_{it} & \text{for} \quad t = \lfloor \tau T \rfloor + 1, \ldots, T, \end{cases}$$

9. Disappearing Factor (Type 2a)

$$X_{it} = \begin{cases} F_t'\lambda_{1i} + e_{it} & \text{for} \quad t = 1, \ldots, \lfloor \tau T \rfloor \\ F_t'\lambda_{2i} + e_{it} & \text{for} \quad t = \lfloor \tau T \rfloor + 1, \ldots, T, \end{cases}$$

The proposed procedure thus far cannot deal with type 2a breaks.

Simulation Results

We present the results of the simulation study.

Unsurprisingly,

Perhaps most interesting is the result of Chen, Dolado, and Gonzalo ((2014))'s test statistic, which seems to exhibit power against Type 2a breaks. However, the distribution of their test statistic under such breaks has not been formally derived.

Empirical Study

To date, there is limited empirical work conducted on the detection of structural breaks in dynamic factor models. However, there exists a strong consensus that structural instabilities in dynamic factor models do exist. For example, James H Stock and Watson ((2021)) show in a seminal study that forecasting using factor models while ignoring potential breaks leads to lower performance, and Breitung and Eickmeier ((2011)), Baltagi, Kao, and Wang ((2021)), Ma and Su ((2018)), suggest evidence in empirical data for the structural break tests they propose. As such, the notion of possibility of structural instabilities has reached the point where many large macroeconomic datasets provide a simple rolling/recursive window study into possible instability, particularly with the changing no. of factors (see McCracken and Ng ((2015)) and McCracken and Ng ((2020)) for evidence in more contemporary US data, Hartigan and Morley ((2020)) for evidence in Australian data, Coulombe, Marcellino, and Stevanović ((2021)) for evidence in UK data, Fortin-Gagnon et al. for evidence in Canadian data).

The lack of precise empirical results is perhaps due to a number of reasons:

1. Possibility of multiple breaks

The majority of the literature thus far has only considered the case of a single break, but the empirical data may be subject to multiple breaks. Although it is well known that single break tests do have power against multiple breaks, the finite sample performance of such tests is notorious for being unreliable, in the sense that they do not always agree which break is the biggest.

2. Possibility of non-smooth breaks

It is also well known that structural break tests may perform poorly when instead of large sudden breaks, the DGP instead exhibits small, smooth changes over time. Su and Wang ((2017)) and Su and Wang ((2020)) show in simulation evidence for this, and propose a non-parametric framework to estimate and test for smooth changes. However, their method tests the null hypothesis of no structural change at all, against the alternative of *any* structural change, and cannot distinguish between smooth changes and sudden changes.

3. Lack of data in the T dimension

The previous two points generally compound the main issue of there not being enough data in each subsample. This issue is profound in the structural break literature - whether using the framework proposed by Andrews ((1993)) or Bai and Perron ((1998)), the resulting test statistics are poorly behaved towards the start and end of the subsample, and hence a proportional of the data, denoted by ϵ, usually around 0.1 or 0.15, is needed to trimmed from both the start and end.

These problems often mean that practically, a practitioner must be careful of making sure that the sample is large enough to avoid boundary issues, but also not large enough that it may include another possible break. As such, the practitioner must have a pre-existing idea of where such breaks may occur already, and "pre-treat" the data accordingly beforehand.

Popular breaks that are typically suggested and/or identified by the literature are:

- Great Moderation, early 1980s
- Dot-com bubble, early 2000s
- GFC, late 2007/early 2008
- COVID pandemic, late 2019/early 2020

It is remarkable that a large amount of the earlier literature has only identified the Great Moderation as a potential break date, but this is due to the data from the 2000s onwards being trimmed out due to lack of data back then.

When these tests are run on more contemporary samples, they tend to also find some of the later break dates.

Data

Stock and Watson 2005 James H. Stock and Watson ((2005)) provides a balanced dataset.

Stock and Watson 2009 James H Stock and Watson ((2021)) provides a balanced dataset.

Stock and Watson 2012 James H. Stock and Watson ((2012)) provides a balanced dataset.

FREDMD McCracken and Ng ((2015)) provides a large, monthly frequency, macroeconomic database based on US data. This is emerged in more recent times to be the *de facto* standard for large empirical macroeconomic databases, and is aimed to be broadly compatible and equivalent to the vintages prepared by Stock and Watson.

We pre-treat the data by removing any series that have missing data. Alternatively, we also consider using imputation via an EM algorithm which uses PCA, and imputation via . All approaches mirror that of McCracken and Ng ((2015)), and in general does not affect the general result much.

We do not do anything special with regards to periods related to COVID19 because it generally falls within the period that is to be trimmed off for structural break tests anyway.

FREDQD McCracken and Ng ((2020)) provides a large, quarterly frequency, macroeconomic database based on US data - the quarterly counterpart to FREDMD.

Empirical Results

Future Work

The work presented thus far has identified notable gap in the literature, namely disentangling

It remains as future work to formally prove and establish the asymptotic validity of the proposed procedure.

It is noteworthy that the procedure bears some resemblance to the Feasible Generalised Principal Components estimator proposed by Choi ((2012)).

VAR approaches to Disentangling Breaks

Han and Inoue ((2015)) in their thesis/working version of their paper do have an empirical section with uses James H Stock and Watson ((2021))'s 2009 dataset.

They find that if the break date of 1984 is known and used a priori, then there is some mixed evidence for there being a break.

However, if the break date itself is unknown and treated as a nuisance parameter, then they cannot reject the null of there being any breaks in the dynamic factor model.

Han and Inoue ((2015)) also make an attempt at disentangling breaks from the factor loadings and factor dynamics/variance, but I do not believe this approach is legitimate, and perhaps why it was omitted from the final published paper.

Essentially, they estimate a VAR for the estimated factors, and run standard VAR based tests. Specifically, this is a test for changes in coefficients and/or variance of residuals in the VAR system. Note that because there are many pseudo factors, instead of a joint test, they instead test the coefficients of each equation and variance of the residuals separately, adn then combine them with a Bonferri correction. They report the results of the test using HAC.

This was also what Ben was hinting towards, i.e. using Qu and Perron type of test.

However, estimation of a VAR requires that the data series be stationary, something which is explicitly not true with pseudo factors, so I am unclear as to how this procedure is valid (it is likely not valid).

Meeting Notes

Type 2a break issue

This is the only way that diappearing/emerging factors are incorporated in the current framework. However, the proposed procedure cannot disentangle this from a type 1 break, which is problematic.

Ben says that this type of break may be ignored for practical purposes, if the empirical data does not show a large amount of evidence for this of break. E.g. a change from 3 factors to 4 factors due to a break does not mean too much, whereas an change from say 3 to 10 factors would be more alarming. Double check on empirical data to see how relevant this is. However, as I recall, the changes tend to be small.

Bonsoo suggested that I look into some related literature based on the singular matrix Z0. Because this matrix is singular, what this would mean is that

Empirical Data Issue

There is some very limited evidence that the proposed correction method is working, at least mildly. There is limited evidence to suggest that the Great Moderation is a change in factor dynamics that is being erroneously picked out, and that the 2000s recession and GFC are "larger" changes where both the factor loadings and factor dynamics have changed. Ben said that this was a promising looking result, but more work would need to be done.

Suggestions: Use the exact same dataset and see what happens.

BE test This consistently finds a break in factor loadings at 1984, for many series. However, as mentioned before, a limitation of this test is that it may be detecting so called small breaks, or mistakenly rejecting the null hypothesis due to the fact that the pseudofactors are being used.

Practical interpretation: the Great Moderation break did indeed induce some breaks in factor loadings for a substantial fraction of series. However, when looking at the entire cross section as a whole, this "break" in facotr loadings is not large enough to constitute a "large" break. Instead, it may be mroe accurate to say that any large break is described as a break in factor dynamics instead.

HI test The working/thesis version of this paper found mixed evidence for the Great Moderation break.

When the break is treated as known a priori, there is mixed evidence. When the break date itself is treated as a nuisance parameter, this evidence falls away.

Proposed Procedure The proposed procedure in general lessens the evidence in support for any kind of break around the Great Moderation.

Timeline

References

Andrews, D. W. K. (1993): "Tests for parameter instability and structural change with unknown change point," *Econometrica*, 61, 821–56.

Bai, J. (2003): "Inferential theory for factor models of large dimensions," *Econometrica*, 71, 135–71.

Bai, J., and X. Han. (2016): "Structural changes in high dimensional factor models," *Frontiers of Economics in China*, 11, 9–39.

Bai, J., X. Han, and Y. Shi. (2020): "Estimation and inference of change points in high-dimensional factor models," *Journal of Econometrics*, 219, 66–100.

Bai, J., and S. Ng. (2002): "Determining the number of factors in approximate factor models," *Econometrica*, 70, 191–221.

---. (2007): "Determining the number of primitive shocks in factor models," *Journal of Business & Economic Statistics*, 25, 52–60.

Bai, J., and P. Perron. (1998): "Estimating and testing linear models with multiple structural changes," *Econometrica*, 66, 47.

Baltagi, B. H., C. Kao, and F. Wang. (2017): "Identification and estimation of a large factor model with structural instability," *Journal of Econometrics*, 197, 87–100.

---. (2021): "Estimating and testing high dimensional factor models with multiple structural changes," *Journal of Econometrics*, 220. Annals issue: Celebrating 40 years of panel data analysis: Past, present and future, 349–65.

Bates, B. J., M. Plagborg-Møller, J. H. Stock, and M. W. Watson. (2013): "Consistent factor estimation in dynamic factor models with structural instability," *Journal of Econometrics*, 177. Dynamic econometric modeling and forecasting, 289–304.

Breitung, J., and S. Eickmeier. (2011): "Testing for structural breaks in dynamic factor models," *Journal of Econometrics*, 163. Factor structures in panel and multivariate time series data, 71–84.

Chen, L., J. J. Dolado, and J. Gonzalo. (2014): "Detecting big structural breaks in large factor models," *Journal of Econometrics*, 180, 30–48.

Choi, I. (2012): "EFFICIENT ESTIMATION OF FACTOR MODELS," *Econometric Theory*, 28, 274–308.

Coulombe, P. G., M. Marcellino, and D. Stevanović. (2021): "CAN MACHINE LEARNING CATCH THE COVID-19 RECESSION?" *National Institute Economic Review*, 256, 71–109.

Duan, J., J. Bai, and X. Han. (2021): "Quasi-maximum likelihood estimation of break point in high-dimensional factor models," *arXiv:2102.12666 [econ]*,.

Forni, M., M. Hallin, M. Lippi, and L. Reichlin. (2000): "The generalized dynamic-factor model: Identification and estimation," *The Review of Economics and Statistics*, 82, 540–54.

Fortin-Gagnon, O., M. Leroux, D. Stevanovic, and S. Surprenant. "A large canadian database for macroeconomic analysis," 52.

Han, X., and A. Inoue. (2015): "TESTS FOR PARAMETER INSTABILITY IN DYNAMIC FACTOR MODELS," *Econometric Theory*, 31, 1117–52.

Hartigan, L., and J. Morley. (2020): "A factor model analysis of the australian economy and the effects of inflation targeting," *Economic Record*, 96, 271–93.

Lam, C., and Q. Yao. (2012): "Factor modeling for high-dimensional time series: Inference for the number of factors," *The Annals of Statistics*, 40.

Lam, C., Q. Yao, and N. Bathia. (2011): "Estimation of latent factors for high-dimensional time series," *Biometrika*, 98, 901–18.

Liu, X., and R. Chen. (2017): "Regime-switching factor models for high-dimensional time series," *Statistica Sinica*,.

---. (2020): "Threshold factor models for high-dimensional time series," *Journal of Econometrics*, 216. Annals issue in honor of george tiao: Statistical learning for dependent data, 53–70.

Liu, X., and T. Zhang. (2022): "Estimating change-point latent factor models for high-dimensional time series," *Journal of Statistical Planning and Inference*, 217, 69–91.

Ma, S., and L. Su. (2018): "Estimation of large dimensional factor models with an unknown number of breaks," *Journal of Econometrics*, 207, 1–29.

McCracken, M. W., and S. Ng. (2015): FRED-MD: A Monthly Database for Macroeconomic Research,.

---. (2020): FRED-QD: A Quarterly Database for Macroeconomic Research,.

Stock, James H., and M. W. Watson. (2005): Implications of Dynamic Factor Models for VAR Analysis, Working Paper, National Bureau of Economic Research.

---. (2012): Disentangling the Channels of the 2007-2009 Recession,National Bureau of Economic Research.

Stock, J. H., and M. W. Watson. (2016): "Chapter 8 - Dynamic Factor Models, Factor-Augmented Vector Autoregressions, and Structural Vector Autoregressions in Macroeconomics," in *Handbook of macroeconomics*, ed. by Taylor, J. B. and H. Uhlig. Elsevier, 415–525.

Stock, James H., and M. W. Watson. (2021): "Forecasting in dynamic factor models subject to structural instability," 65.

Su, L., and X. Wang. (2017): "On time-varying factor models: Estimation and testing," *Journal of Econometrics*, 198, 84–101.

---. (2020): "TESTING FOR STRUCTURAL CHANGES IN FACTOR MODELS VIA a NONPARAMETRIC REGRESSION," *Econometric Theory*, 36, 1127–58.

Yamamoto, Y., and S. Tanaka. (2015): "Testing for factor loading structural change under common breaks," *Journal of Econometrics*, 189, 187–206.